(iii) $V(x) = E[x^2] - [E(x)]^2$

$V(x) = E[x^2] - E(x)]^2$

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Prob: Given,

$$f(x) = k(x+1) \; ; \; 2 \leq x \leq 5$$

(i) Find the value of k

(ii) $P(x > 3)$

(iii) $P(x < 4)$

(iv) $E[x]$

(v) $V(x)$

Soln: (i) We know,

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

$$\Rightarrow \int_{2}^{5} k(x+1)\, dx = 1$$

$$\Rightarrow k\left[\frac{x^2}{2} + x\right]_2^5 = 1$$

$$\Rightarrow k\left[\frac{1}{2}(5^2 - 2^2) + (5-2)\right] = 1$$

$$\Rightarrow k\left[\frac{27}{2}\right] = 1$$

$$\Rightarrow k = \frac{2}{27}$$

therefore,

$$f(x) = \frac{2}{27}(x+1) \; ;$$

$$2 \leq x \leq 5$$

(ii) $P(x>3) = \int_{3}^{5} f(x)\,dx$

$= \int_{3}^{5} \frac{2}{27}(x+1)\,dx$

$= \frac{2}{27}\left[\frac{x^2}{2} + x\right]_{3}^{5} dx$

$= \frac{2}{27} \times \frac{1}{2}\left[(5^2 - 3^2) + (5-3)\right]$

$= \frac{2}{54}\left[16 + 2\right]$

$= \frac{2}{54} \times 18 = \frac{2}{3}$

(iii) $P(x<4) = \int_{2}^{4} \frac{2}{27}(x+1)\,dx$

$\Rightarrow \frac{2}{27}\int_{2}^{4}(x+1)\,dx$

$\Rightarrow \frac{2}{27}\left[\frac{x^2}{2} + x\right]_{2}^{4}$

$\Rightarrow \frac{2}{27}\left[\frac{1}{2}(4^2 - 2^2) + (4-2)\right]$

$\Rightarrow$

(iv) $E[x] = \int_{-\infty}^{\infty} x f(x)\, dx$

$= \int_{2}^{5} x \cdot \frac{2}{27}(x+1)\, dx$

$= \frac{2}{27} \int_{2}^{5} x^2 + x\, dx$

$= \frac{2}{27} \int_{2}^{5} \left(\frac{x^3}{3} + \frac{x^2}{2}\right) dx$

$= \frac{2}{27} \times \frac{1}{3} \times \frac{1}{2} \left[(5^3 - 2) + (5^2 - 2)\right]$

$= \frac{1}{81}\left[23 + 123\right]$

$= \frac{1}{81} \times 146 = 1.80$

(v) $V(x) = E[x^2] - [E(x)]^2$

$= \int_{-\infty}^{\infty} x^2 f(x)\, dx - [E(x)]^2$

$= \int_{2}^{5} x^2 \cdot \frac{2}{27}(x+1)\, dx - [E(x)]^2$

$= \frac{2}{27} \int_{2}^{5} x^3 + x^2\, dx - [E(x)]^2$

Prev. 4 sem এর এই ডিফারেন্ট Math করো।

$f(x) = k(x-1)$; $2 \leq x \leq 6$.

Ch-4: Correlation & Regression:

1. What is Correlation?

⇒ Consider the data:

x	y
1	6
2	4
	3

$\sum x = 21$, $\sum y = 24$,

$\sum x^2 = 91$ $\sum y^2 = 10$